Numerical Differentiation

Chapter 02

Lesson 2

Backward Difference Formulas

Learning Objectives

At the end of this lecture, students should be able to:

- Explain the concept of numerical differentiation.
- Define the backward difference operator.
- Construct a backward difference table.
- Derive backward difference formulas using Taylor series.
- Apply first-, second-, and higher-order backward difference formulas.
- Calculate truncation errors.
- Implement backward difference formulas using MATLAB.
- Identify when backward difference formulas are appropriate.
- Compare forward and backward difference formulas.

1. Finite Difference Concept

- A **finite difference** approximates the change in a function between discrete points.

Suppose the values of x are equally spaced:

$$x_0, x_1, x_2, \ldots$$

Where,

$$x_{i+1} - x_i = h.$$

Therefore,

$$x_{i-1} = x_i - h$$
$$x_{i-2} = x_i - 2h$$
$$x_{i-3} = x_i - 3h.$$

- For backward difference formulas, we use function values at the **current point and points to its left**.
- For example, to estimate $f'(x_i)$, we can use

$$f(x_i), f(x_i - h), f(x_i - 2h), \ldots..$$

This is the main difference between forward and backward difference methods.

Suppose we are currently at x_{i} .

- Because the points are separated by h, the point immediately to the **left** of xi is:

$$x_{\mathrm{i}-1} = x_{\mathrm{i}} - h$$

- For example, suppose

$$x_{\mathrm{i}} = 5, \qquad \text{h=1.}$$

Then:

$$x_{\mathrm{i}-1} = 5-1=4.$$

- So xi−h means **one step backward**.

What does $x_i - 2h$ mean?

- If we go two steps backward from xi:

$$x_{i-2} = x_i - 2h$$

- For example:

$$x_i = 5,\ h = 1$$

then

$$x_{i-2} = 5 - 2(1) = 3$$

So:

$x_i - 2h$ =two steps to the left

Similarly,

$$x_{i-3} = x_i - 3h$$

means three steps to the left

Visualize the points

Suppose h is constant:

$$\cdots \quad x_i - 3h \quad x_i - 2h \quad x_i - h \quad x_i \quad x_i + h \quad x_i + 2h \quad \cdots$$

Think of x_i as the **current point.**

- The **forward difference** looks to the **right** of the current point.
- The **backward difference** looks to the **left** of the current point.

The **backward difference operator** is denoted by

$$\nabla$$

The first backward difference is defined as

$$\nabla f(x) = f(x) - f(x - h)$$

For tabulated data,

$$\nabla f_i = f_i - f_{i-1}$$

Therefore,

The simplest backward difference formula for the first derivative is

$$f'(x_i) \approx \frac{f_i - f_{i-1}}{h}$$

or

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}$$

This is called the **first-order backward difference formula.**

Why is it called "backward"?

Because from x_i, we move **backward**:

$$x_i \longrightarrow x_i - h$$

For example:

$$x_i - 2h \qquad \underbrace{x_i - h}_{\text{previous point}} \qquad \underbrace{x_i}_{\text{current point}}$$

The derivative is estimated using the current point and the point behind it.

Forward vs backward

Method	Points used	First difference
Forward	$x_i,\ x_i + h$	$f(x_i + h) - f(x_i)$
Backward	$x_i - h,\ x_i$	$f(x_i) - f(x_i - h)$

$$f'(x_i) \approx \frac{f(x_i + h) - f(x_i)}{h}$$

$$f'(x_i) \approx \frac{f(x_i) - f(x_i - h)}{h}$$

Why is backward difference useful?

Imagine you are at the last point in a dataset:

$$x_0, \quad x_1, \quad x_2, \quad x_3$$

Suppose you want $f'(x_3)$.

There is no x_4 available.

Therefore, you cannot easily use

$$f(x_3 + h).$$

But you do have previous points:

$$x_2, \quad x_1, \quad x_0.$$

So you can use the **backward difference**:

$$\boxed{f'(x_3) \approx \frac{f(x_3) - f(x_2)}{h}}$$

This is one of the main reasons backward difference formulas are useful.

Derivation of First-Order Backward Difference Formula

To derive the formula, use the Taylor expansion of

$$f(x-h)$$

about x.

Taylor series gives

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2!}f''(x) - \frac{h^3}{3!}f'''(x) + \cdots$$

Therefore,

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(x) + \cdots$$

Rearranging,

$$f(x) - f(x-h) = hf'(x) - \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) - \cdots$$

Divide by h:

$$\frac{f(x) - f(x-h)}{h} = f'(x) - \frac{h}{2}f''(x) + \frac{h^2}{6}f'''(x) - \cdots$$

Therefore,

$$\boxed{f'(x) \approx \frac{f(x) - f(x-h)}{h}}$$

The truncation error is

$$\boxed{O(h)}.$$

Example 1:

Given

$$f(x) = x^2 + 2x$$

estimate

$$f'(1)$$

using

$$h = 0.1.$$

Step 1: Formula

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}$$

Therefore,

$$f'(1) \approx \frac{f(1) - f(0.9)}{0.1}.$$

Step 2: Calculate $f(1)$

$$f(1) = 1^2 + 2(1)$$

$$= 3.$$

Step 3: Calculate $f(0.9)$

$$f(0.9) = (0.9)^2 + 2(0.9)$$

$$= 0.81 + 1.8$$

$$= 2.61.$$

Step 4: Substitute

$$f'(1) \approx \frac{3 - 2.61}{0.1}$$

$$= \frac{0.39}{0.1}$$

$$\boxed{f'(1) \approx 3.9}$$

Exact answer

$$f'(x) = 2x + 2.$$

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$$f'(x) = 2x + 2.$$

Therefore,

$$f'(1) = 4.$$

Hence,

$$\boxed{\text{Absolute error} = |4 - 3.9| = 0.1}$$

Second-Order Backward Difference Formula

- The second-order backward difference is obtained by applying the backward difference operator twice.

The first backward difference is

$$\nabla f_i = f_i - f_{i-1}.$$

The second backward difference is

$$\nabla^2 f_i = \nabla(\nabla f_i).$$

Therefore,

$$\nabla^2 f_i = \nabla f_i - \nabla f_{i-1}.$$

Now,

$$\nabla f_i = f_i - f_{i-1}$$

and

$$\nabla f_{i-1} = f_{i-1} - f_{i-2}.$$

Therefore,

$$\begin{aligned}\nabla^2 f_i &= (f_i - f_{i-1}) - (f_{i-1} - f_{i-2}) \\ &= f_i - 2f_{i-1} + f_{i-2}.\end{aligned}$$

Thus,

$$\boxed{\nabla^2 f_i = f_i - 2f_{i-1} + f_{i-2}}$$

Backward Difference Formula for the Second Derivative

The second derivative can be approximated by

$$\boxed{f''(x_i) \approx \frac{f_i - 2f_{i-1} + f_{i-2}}{h^2}}$$

or

$$\boxed{f''(x) \approx \frac{f(x) - 2f(x-h) + f(x-2h)}{h^2}}$$

The truncation error is

$$\boxed{O(h)}.$$

Example:

Consider

$$f(x) = x^3 + 2x^2 + x.$$

Estimate

$$f''(1)$$

using

$$h = 0.1.$$

Step 1: Formula

$$f''(1) \approx \frac{f(1) - 2f(0.9) + f(0.8)}{(0.1)^2}.$$

Step 2: Calculate $f(1)$

$$f(1) = 1 + 2 + 1 = 4.$$

Step 3: Calculate $f(0.9)$

$$f(0.9) = (0.9)^3 + 2(0.9)^2 + 0.9$$

$$= 0.729 + 1.62 + 0.9$$

$$= 3.249.$$

Step 4: Calculate $f(0.8)$

$$f(0.8) = (0.8)^3 + 2(0.8)^2 + 0.8$$

$$= 0.512 + 1.28 + 0.8$$

$$= 2.592.$$

Step 5: Substitute

$$f''(1) \approx \frac{4 - 2(3.249) + 2.592}{0.01}$$

$$= \frac{4 - 6.498 + 2.592}{0.01}$$

$$= \frac{0.094}{0.01}$$

$$\boxed{f''(1) \approx 9.4}$$

Exact answer

$$f''(x) = 6x + 4.$$

Therefore,

$$f''(1) = 10.$$

Hence,

$$\boxed{\text{Absolute error} = |10 - 9.4| = 0.6}$$

Higher-Order Backward Difference Formulas

- Higher-order backward differences are obtained by repeatedly applying the backward difference operator.

First Backward Difference

$$\nabla f_i = f_i - f_{i-1}$$

Second Backward Difference

$$\nabla^2 f_i = f_i - 2f_{i-1} + f_{i-2}$$

Third Backward Difference

$$\nabla^3 f_i = f_i - 3f_{i-1} + 3f_{i-2} - f_{i-3}$$

Fourth Backward Difference

$$\nabla^4 f_i = f_i - 4f_{i-1} + 6f_{i-2} - 4f_{i-3} + f_{i-4}$$

General Backward Difference Formula

The general n-th backward difference is

$$\nabla^n f_i = \sum_{k=0}^{n} (-1)^k \binom{n}{k} f_{i-k}$$

For example, for $n = 3$:

$$\nabla^3 f_i = \sum_{k=0}^{3} (-1)^k \binom{3}{k} f_{i-k}$$

Therefore,

$$\nabla^3 f_i = f_i - 3f_{i-1} + 3f_{i-2} - f_{i-3}.$$

Backward Difference Table

Consider:

x	$f(x)$
0	1
1	4
2	9
3	16
4	25

The backward difference table can be written as:

x	$f(x)$	∇f	$\nabla^2 f$	$\nabla^3 f$
0	1			
1	4	3		
2	9	5	2	
3	16	7	2	0
4	25	9	2	0

Question

Consider the following data:

x	1	2	3	4	5
$f(x)$	2	8	18	32	50

Construct the backward difference table by calculating:

1. The first backward differences, ∇f.
2. The second backward differences, $\nabla^2 f$.
3. The third backward differences, $\nabla^3 f$.

Hence, determine whether the given data represent a quadratic relationship.

First Backward Differences

The formula is

$$\boxed{\nabla f_i = f_i - f_{i-1}}$$

Therefore,

$$\nabla f_1 = 8 - 2 = 6$$

$$\nabla f_2 = 18 - 8 = 10$$

$$\nabla f_3 = 32 - 18 = 14$$

$$\nabla f_4 = 50 - 32 = 18$$

Second Backward Differences

The formula is

$$\boxed{\nabla^2 f_i = \nabla f_i - \nabla f_{i-1}}$$

Thus,

$$\nabla^2 f_2 = 10 - 6 = 4$$

$$\nabla^2 f_3 = 14 - 10 = 4$$

$$\nabla^2 f_4 = 18 - 14 = 4$$

Third Backward Differences

$$\nabla^3 f_i = \nabla^2 f_i - \nabla^2 f_{i-1}$$

Therefore,

$$\nabla^3 f_3 = 4 - 4 = 0$$

$$\nabla^3 f_4 = 4 - 4 = 0$$

Complete Backward Difference Table

x	$f(x)$	∇f	$\nabla^2 f$	$\nabla^3 f$
1	2			
2	8	6		
3	18	10	4	
4	32	14	4	0
5	50	18	4	0

The third backward differences are zero:

$$\boxed{\nabla^3 f = 0}$$

This indicates that the given data are generated by a **quadratic function.**

In fact, the data come from

$$\boxed{f(x) = 2x^2}$$

because

$$f(1) = 2, \quad f(2) = 8, \quad f(3) = 18, \quad f(4) = 32, \quad f(5) = 50.$$

This is a good example to show students how **constant second differences indicate a quadratic relationship.**